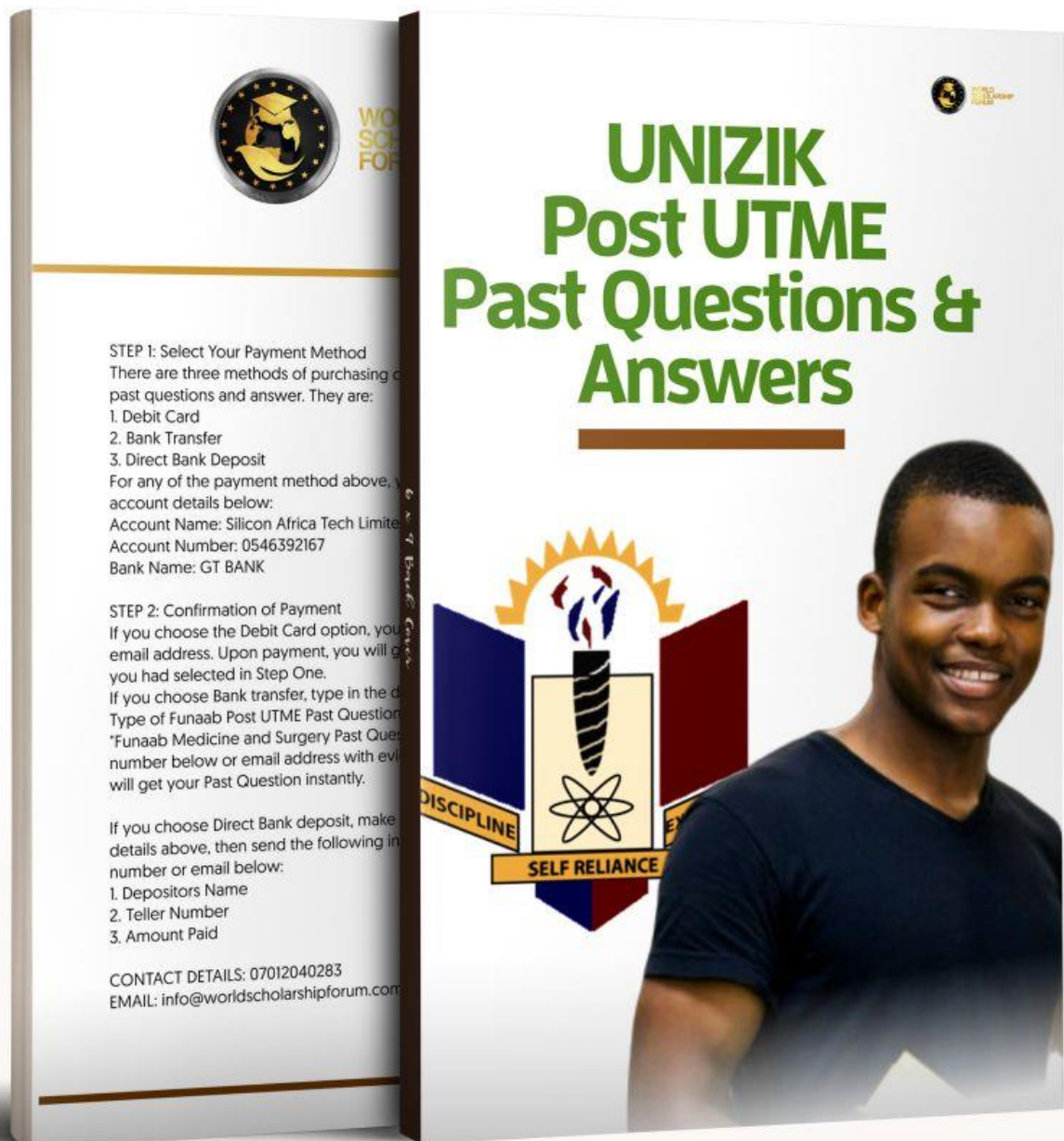
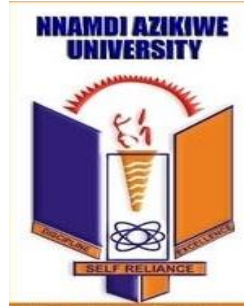






POST-UME SCREENING EXERCISE, 2011/2012 FOR HEALTH SCIENCES AND TECHNICAL MEDICINE, PHARMACY AND NATURAL SCIENCES

Shade only one answers using HB pencil only Time allowed: 1 hour

INSTITUTIONS

Choose the option that best conveys the meaning of the underlined portion in each of the following sentences

1. In the match against the Rangers team, the Enyimba boys turn out to be the dark horse. (A) played most brilliantly (B) played below their usual form (C) Won unexpectedly (D) lost as expected (E) Won as expected.
2. He spoke with his heart in his mouth. (A) Courageously (B) With such unusual cowardice (C) With a lot of confusion in his speech (D) Without being able to make up his mind (E) With fright and agitation.
3. The leader in today's issue of Punch Newspaper focuses in inflation. (A) President (B) Heading (C) Editorial (D) Columnist (E) Proprietor.
4. All his plans fell through. (A) failed (B) Were accomplished (C) Had to be reviewed (D) Were rejected (F) Fell.
5. He stared at her. (A) Glance (B) Peeped (C) Looked (D) Looked (E) Gazed (F) Fixed.
6. As he was a gullible leader, his followers took advantage of him. (A) He was weak and unable to enforce his authority (B) He was partial and unfair in dispensing justice (C) He was simple minded to a fault (D) He was slow to act (E) He was as lacking in education and experience in everyday affairs.
7. His summary of the lecture was brief and to the point. (A) Careful (B) Precise (C) Accurate (D) Exact (E) Crucial.
8. Do you have the same aversion as I do home videos? (A) Bitterness (B) Dislike (C) Criticism (D) Indignation (E) Preference.
9. I did not think she could be so easily, taken in by his pretence (A) Flattered (B) Deceived (C) Pregnant (D) Overcome (E) Blackmailed.
10. The witness was guilty of swearing to a statement he knew to be false. (A) Forgery (B) Perjury (C) Libel (D) Slander (E) Affidavit.
11. Most of his observations were wide of the mark. (A) Comprehensive (B) Irrelevant (C) Pertinent (D) Came rather late (E) Unaccepted.
12. The clerk refused to answer for the mistakes made by manager and his assistants. (A) To reply to (B) To give an answer (C) To explain (D) To accept responsibility (E) To account.
13. Good citizens should take the part of government when it is being attacked by the foreign press. (A) Speak in support of (B) Explain the part played by (C) Act like a representative of (D) Emphasize the progress made by (E) Outline the virtues of.
14. The Principal of my school has a variety of peculiarities. The most conspicuous is the way he gesticulates when making a serious point. (A) Shows anxiety when making a serious point (B) Breaks down when talking seriously (C) Uses his hands and arms as a sign (D) Bows his head in humility (E) Shakes his head from one side to the other.
15. He heard the loud noise of the huge air-raid siren. (A) Buzz (B) Crack (C) Rumble (D) Whistle (E) Wail.
16. One of the stages of the creative process is the incubation period. This refers to the period when. (A) The idea is firmly fixed through repetition (B) The person suddenly comes upon the idea (C) The persons thinks over the idea (D) The idea yields concrete results (E) The idea given up.
17. When a man is immune to an illness, he is (A) Opposed to it (B) Attached to it (C) Hated by it (D) Protected against it (E) Addicted to it.
18. The two boxers were neck and neck up till the sixth round but in the seventh and final round, the bigger one lost ground. (A) Could no longer stand on his feet (B) Failed to maintain his position (C) Was beaten (D) Became dizzy (E) Became a punching bag.
19. He uses his savings to bring up his nephew (A) Lead (B) Accompany (C) Feed (D) Support (E) Conduct
20. Whenever I travel abroad, I always bring on my return, photographs, postcards and other small items as souvenirs. (A) Mementoes (B) Remembrance (C) Memorials (D) Memorable (E) Property.

INSTRUCTIONS

COMPLETE EACH OF THE FOLLOWING SENTENCES BY CHOOSING THE

OPTION THAT MOST SUITABLY FILLS THE SPACE

21. When the beggar was tired he _____ down by the roadside. (A) Lied (B) Laid (C) Layed (D) Lay (E) Lain.
22. He did not like _____ leaving the class early. (A) We (B) Us (C) Our (D) Ourselves (E) Ourselves.
23. Before the operation the dentist found that his patient's teeth _____ (A) Have long decayed (B) Have long been decayed (C) Have long being decayed (D) had long decayed (E) Had been decayed.
24. The employer, not the salesman and his representative responsible for the loss of revenue.

25. The boy was born before his parents actually got married and so the court has declared him _____ (A) Illegal (B) Illegitimate (C) Illicit (D) Unlawful (E) Untenable.
26. As he was still in me two naira. I was careful not to _____ him any money. (A) Credit (B) Loan (C) Borrow (D) Extend (E) Lend.
27. Last week the _____ at the theatre was the largest I had ever seen. (A) Assembly (B) Congregation (C) Convention (D) Audience (E) Crowd
28. The green snake which lives in the green grass is using a perfect _____ (A) Ambush (B) Undercover (C) Camouflage (D) Shelter (E) Subterfuge
29. In order to carry out the necessary post mortem examination, the dead body was _____ (A) Extracted (B) Extradited (C) Exiled (D) Exhumed (E) Expelled
30. Remember this matter is strictly _____ (A) Between you and I (B) Among you and .1 (C) Among you and me (D) Between you and me. (E) Between you and myself
31. Equilibrium is said to be attained in a reversible reaction when A. All the reactions have been used up B. All the products have been formed C. There is no further change in temperature D. The rates of the forward and backward reactions are equal E. The rate of formation of the products decreases with time
32. Bubbling of carbon (IV) oxide into calcium hydroxide solution results in the formation of (A) CaO (B) Ca(OH)₂ (C) CaCO₃ (D) Ca(HCO₃)₂ (E) H₂CO₃
33. If 3 mole of electrons are required to deposit 1 mole of a metal M during the electrolysis of its molten chloride, the empirical formula of the metallic chloride is (A) M₂Cl (B) M₃Cl (C) MCl (D) M₂Cl₃ (E) MCl₃
34. Which of the following compounds crystallizes without water of crystallization? (A) Na₂CO₃ (B) CuSO₄ (C) MgSO₄ (D) NaCl (E) FeSO₄
35. When starch undergoes complete enzyme catalyzed hydrolysis, the resulting product is (A) Glucose (B) Maltose (C) Sucrose (D) Fructose (E) Cellulose.
36. When steam is passed over white-hot coke the products are. (A) Carbon (IV) oxide and nitrogen (B) Carbon (IV) oxide and hydrogen (C) Carbon (II) oxide and nitrogen (D) Carbon (IV) oxide and hydrogen (E) Carbon (IV) oxide and steam.
37. Which of the following compounds will not decompose when heated strongly? (A) NaHCO₃ (B) K₂CO₃ (C) MgCO₃ (D) Ca(HCO₃)₂ (E) ZnCO₃.
38. The rate of a reaction is proportional to the number of effective collisions accounting per second between the reactants, this statement is associated with the. (A) Kinetic theory (B) Rate law (C) Atomic theory (D) Collision theory (E) Gas laws.
39. How many faradays of electricity are required to liberate 9g of aluminum? (Al = 27). (A) 0.1 (B) 0.3 (C) 1.0 (D) 2.7 (E) 3.0.
40. Which of the following metals will be the most suitable for use where lightness and resistance to corrosion are of importance? (A) Lead (B) Copper (C) Iron (D) Calcium (E) Aluminum.
41. If you cross-breed a tall variety (Tt) of maize with a short variety (tt), the ratio of tall to short plants in the offspring will be. (A) 2:1 (B) 1:2 (C) 1:1 (D) 1:3 (E) 3:1.
42. Euglena may be classified as a plant because it. (A) has chloroplast (B) Has a gullet (C) Lives in a pond (D) Possess a flagellum (E) Has a pellicle.
43. When proteins are broken down they also provide. (A) Oxygen (B) Carbohydrates (C) Energy (D) Amino acids (E) Carbon dioxide.
44. Which of these products obtained from analyzing sweat from the skin is not an excretory product? (A) Water (B) Salts (C) Dust (D) Urea (E) Urates.
45. The centre which controls respiratory activities in the mammalian brain is the. (A) Cerebrum (B) Olfactory lobes (C) Pituitary organ (D) Cerebellum (E) Medulla oblongata.
46. The removal of a man's pancreas by surgical operation can affect only the digestion of. (A) Starch (B) Starch, proteins and fats (C) Oils and fats (D) Proteins (E) Carbohydrates and fats.
47. Which of the following does a virus have in common with animal cells? (A) Nucleus (B) DNA (C) Glycogen (D) Cytoplasm.
48. The sex of a foetus is determined during. (A) Meiosis (B) copulation (C) Fertilization (D) Placentation.
49. The organ located within the duodenal loop in the mammal is the. (A) spleen (B) pancreas (C) Liver (D) Gall bladder
50. Where is energy produced in a. (A) nucleus (B) Nucleolus (C) Lysosomes (D) Chloroplast (E) Mitochondria

**ANSWERS TO NAU POST-UME SCREENING EXERCISE (2011/2012) SESSION
HEALTH SCIENCE / TECHNOLOGY, MEDICAL, PHARMACY AND NATURAL SCIENCES**

1. C.	2. E	3. C	4. A	5. D	6. A	7. D	8. B	9. 8	10. C	11. E	12. D
13. A	14. C	15. D	16. C	17. D	18. B	19. D	20. A	21. E	22. B	23. B	24. E
25. B	26. E	27. D	28. C	29. D	30. A	31. D	32. D	33. C	34. D	35. A	36. D
37. B	38. D	39. C	40. E	41. C	42. A	43. C	44. C	45. E	46. C	47. B	48. C
49. B	50. E										

NNAMDI AZIKIWE UNIVERSITY, AWKA



POST-UME SCREENING EXERCISE, 2012/2013

FOR

AGRICULTURE, HEALTH SCIENCE AND TECHNOLOGY, MEDICINE, NATURAL SCIENCES AND PHARMACEUTICAL SCIENCES

TIME ALLOWED: 1. HOUR DATE AUGUST

Shade (FIR MLY) only one answer using HB pencil only

Please shade the following corrected subject codes

1. Pharmaceutical sciences: 654 and NOT 704
2. Statistics: 554 and NOT 254
3. Agriculture 714

Read the passage carefully and answer the questions that follow it

So far, I have been speaking of science in its university, viewed from the perspective of the world at large. For in the context of our own country and our sister developing countries, many of the factors mentioned earlier are not very important. For example, pollution, deterioration of the environment and population expulsion are not yet serious problems for its in this country.

Let me turn to a more specific area, namely, the question of scientific choice for developing countries. There is no doubt that the role which science and technology have played in the upliftment of the materials and economics well being of the developed nations will, and does, influence the criteria that the third world nations must choose in order to establish their science policies and priorities.

But the criteria to be used by these nations do not have to be the same as those which have brought the developed countries to their present stage of evolution. For while human beings have the same problems, their solutions, to be meaningful will have to be sought within some relevant frame of reference, such as the available resources and expertise, social values, place and time in the historical scale.

1. "Frame of reference" in this passage means. A. Point of information B. range of authority C. economic pattern D. Context of situation
2. The writer apparently believes that through science and technology our country can achieve A. Peace and order B. Progress and material success C. Social values and universality D. Wealth and economic progress
3. According to the passage, the basic consideration for developing science and technology should be three of the following
 1. Technical know-how
 2. Availability of raw materials
 3. Atmospheric pollution
 4. The people's traditions and beliefs
 5. Population
 6. CapitalA. 2.1 and 6 only B. 3.2 and 4 only C. 5.3 and 6 only D. 5.2 and 6 only.
4. The main idea of this passage is that. A. There is a yawning gap between the haves and have-nots B. There could be atmospheric pollution and population explosion

- C. The concerns of science and technology are the same in all countries D. Each nation must plan its development according to its needs and resources.
5. The term "our sister developing countries" implies. A. Britain, Gambia and Canada B. Ghana, Germany and Togo C. China, India and Holland D; Liberia, Sierra-Leone .and Gabon

Choose the expression or word which best completes the sentence

6. _____ for the term for six years, he decided to join another A. when he played B. playing C. having played D. when he was playing.
7. The bridge connecting the two cities was _____ by the enemy A. blown up B. blown down C. having played D. when he was playing.
8. My little boy is suffering _____ jaundice A. from B. of C. for D. with.
9. They went to the market and bought a suitcase and _____ bag A. a big leather brown B. a leather brown big C. a big brown leather D. a brown big leather.
10. By the end of this year, _____ in this town for eleven years. A. I'm living B. I'd be living C. I'll have lived D. I have lived.
11. If he had left home earlier, he _____ late. A. would not be B. would have not been C. would have been not D. would not have been.
12. I saw you walking in that direction but I did not know exactly _____ A. where are you going B. where you were going C. to where you were going D. where you are going.
13. The hill behind the town was so steep. A. That no man could find a foot-hold on b. For no man to find a foothold on c. For any man to find a foot-hold on d. That no man could find a foot-hold on it.
14. Walking down the Street _____ A. the boy was knocked down by a car B. a car knocked the boy down C. the boy was knocked down by car D. a car collided with the boy.
15. The project which seemed very near realization _____ because of lack of funds. A. Fell over B. fell through C. fell in D. fell off.

16. The Chairman of the company was very sympathetic _____ his staff A. to B. at C. for D. on.
17. Jane asked John _____ A. Whether could she meet him later b. Could she meet him later c. If she could meet him later d. She could meet him later
18. She doesn't have to attend the lecture _____ A. doesn't B. is she? C. has she? D. does she?
19. After a series of quarrels, it would be _____ to pretend that I have any more regard for him. A. hypothetical B. essential C. hypocritical D. inevitable.
20. If the company had any delay at the customs office, it was _____ their own making A. at B. of C. from D. through.
21. The members of the panel were _____ A. discussing about it B. discussing on it C. discussing it D. discussing around it

Choose the word/ expression which best completes each sentence

22. The giant hydro-electric project is among the _____ of colonial rule in Southern Africa A. inheritance B. remnants C. legacies D. evidence.
23. During the inaugural address, the president _____ the activities of his government for the past eight years. A. reiterated B. reviewed C. restated D. recited.
24. The World Bank team which visited the landlocked country _____ a bleak economic future for it A. forecast B. discovered C. thought of D. prophesied.
25. We traveled all night and arrived _____ the Lagos motor park at six o'clock in the morning A. by B. inside C. in D. at.
26. We ought to stay away from this place _____ the robbers come back A. provided B. should in case C. in case D. since.
27. The principle will be going away on _____ in his absence, the vice-principal will _____ the school. A. holidays/take over B. holiday/carry on C. holidays/care for D. holiday/look after.
28. The new leader has good intentions, but he is unable to _____ A. carry them with him B. carry them out C. carry them on D. carry them well
29. Enugu is not far from here, it is only an _____ A. hour of driving B. hour's drive C. hour by driving D. hours drive.
30. Do you know if the new lecturer _____ yet? A. is arrived B. will be arriving C. has arrived D. had arrived.
31. Had he known he _____ away A. will go B. will be going C. would have gone D. will have gone.
32. It is time we _____ A. should be going B. ought to go C. go D. went.
33. He sent the children to play _____ he might be alone A. because B. so that C. since D. while.
34. The frictional force between two bodies. A. Exists only when there is relative motion or tendency for motion B. Acts so as to oppose the motion C. Depends on the normal reaction between the two surfaces D. Has all of these characteristics
35. An ice cube floats in a glass of water filled to the brim. What happens when the ice melts? A. The water level remains the same B. The water in the glass overflows
36. There is a drop in the level of water in the glass due to condensation of its outside D. The levels of water drops because melted ice occupies less volume
36. The internationally agreed system of units (SI) for physical measurements A. lb, ft, sec B. g, m, sec C. kg, m, sec D. kg, cm, sec.

37. To determine the weight of an object, you could: A. use a beam balance B. use a spring balance C. find the force necessary to give it a certain acceleration D. use none of these methods.
38. The density of a solid is defined as the A: mass of the solid compared to the mass of an equal volume of water B. amount of water displaced when a unit mass of the solid is immersed in it C. volume per unit mass of the solid D. mass per unit volume of the solid.
39. A few grains of table salt were put in a cup of cold water kept at a constant temperature and left undisturbed. Eventually, all the water tasted salty, this action is due to A. convection B. osmosis C. capillarity D. Diffusion.
40. The heat from the sun reaches the earth by the process of A. insulation B. conduction C. convection D. radiation.
41. Two lamps rated 40w and 220v each are connected in series. The total power dissipated in both lamps is; A. 10w B. 20w C. 40w D. 80w.
42. The amount of heat required to raise the temperature of a body is: A. thermal energy B. thermal conduction C. specific heat capacity D. thermal capacity.
43. If PHCN charges N5 per KWh, what is the cost of operating for 24 hours a lamp requiring 1 A on a 200v line? A. N24 B. N32 C. N55 D. N40.
44. The electric supply from the national grid at Kainji dam is transmitted at high voltages of the order of hundreds of kilovolts but it is converted to domestic use of about 200 to 220 volts by: A. commutators B. transistors C. valves and diodes D. Transformers.
45. Two boys are communicating with each other by stretching a string passing through a hole punched in the bottom of each of two tins. The physical principle employed is that sound travels. A. mainly through air B. faster through gases than in solids and liquids C. with greater ease through a string than in air D. none of the above is correct.
46. The temperature at which the water vapour present in the air saturated the air and begins to condense is known as A. boiling point B. melting point C. critical temperature D. dew point.
47. An object is positioned between two plane mirrors inclined at right angles to each other. The object is 1 unit distance from each mirror. The number of images formed is: A. 1 B. 2 C. 3 D. 4.
48. If the volume of a given mass of gas at constant pressure is proportional to the absolute temperature, the gas is said to obey: A. Hooke's law B. Boyle's law C. Dalton's law D. Charles's law.
49. Which of the following does not cause a reduction of the surface tension of water? A. Detergent B. Alcohol C. Camphor D. Grease.
50. A generator is connected to fifteen 40w lamps and a musician's 600w amplifier. How much energy is used if the generator runs for 6 hours? A. 8.40kwh B. 7.20kwh C. 12.00kwh D. 6.00kwh.
51. Which of the following elements is the best absorber of X-rays? A. Hydrogen B. Oxygen C. Copper D. Lead.
52. When heat is applied to one end of a metal rod, the molecules at the other end soon begin to vibrate with greater amplitude than before because heat has been transferred by A. radiation B. convection C. conduction D. insulation.
53. The plane mirrors in a Kaleidoscope are usually placed A. at an angle of 60° B. parallel to one another C. perpendicular to one another D. at an angle of 45°.

54. Which of the following statement explains why tetraoxosulphate (VI) acid is regard as a strong acid? A. the acid is concentrated B. the acid is highly corrosive C. the acid is completely ionized D. Tetraoxosulphate (VI) ions are very reactive.
55. Which of the following is not correct of glucose and fructose? A. they are structural isomers B. they reduce fehling's solution C. they elm be obtained by the hydrolysis starch D. they are readily fermented by enzyme
56. Which of the following is normally used for drying ammonia gas? A. concentrated retraoxoxuphate (VI) acid B. calcium oxide C. Anhydrous calcium chloride D. phosphorus (V) oxide.
57. Nuclear reactions can be used in the following except. A. gauging the thickness of objects B. making atomic bombs C. generating electricity D. purifying water.
58. The following metals are extracted by the electrolytic method except A. potassium B. Tin C. calcium D. magnesium.
59. PMS (petrol) can be obtained from AGO (diesel) by A. distillation B. cracking C. catalysis D. Polymerization.
60. The separation of a mixture of calcium trioxcarbonate (IV) and sodium trioxcarbonate (IV) is most easily carried out by using the differences in their A. physical states B. melting points C. solubility D. boiling points.
61. What is the amount (in mole) of sodium trioxcarbonate (IV) in 5.3g of the compound? (Na = 23; C = 12; O = 16) A. 0.05 B. 0.10 C. 0.50 D. 0.20.
62. Brass is an alloy of A. Copper and Iron B. copper and chromium C. copper and zinc D. copper and magnesium.
63. The products of the electrolysis of dilute sodium chloride solution with platinum electrodes are A. hydrogen and oxygen B. oxygen and chlorine C. oxygen and water D. chlorine and water.
64. When starch undergoes complete enzyme —catalyzed hydrolysis, the resulting product is A. glucose B. Maltose C. sucrose D. Fructose.
65. Which of the following is not a direct petroleum product> A. Ethanol B. Methane C. Vaseline D. Kerosene.
66. Which of the following properties is not characteristic of electrovalent compound? A. formation of crystalline solids B. solubility in polar solvents C. High vapour pressure D. High melting and boiling points.
67. Compound that have the same molecular formula but different structures are said to be A. Allotropic B. polymeric C. isomeric D. Isotopic.
68. Water from a river contaminated by sodium hydroxide waste will have a PH of about A.L B.3 C.7 D.9.
69. Which of the following is suitable for determining different isotopes present in an element which exhibits isotopy? A. Isotope counter B. cathode 4ay tube C. Mass spectrometer D. Geiger Muller counter.
70. When iron rusts, it undergoes A. deliquescence B. chemical decomposition C. hydrolysis D. redox reaction.
71. When sucrose is warned with fehling's solution A. the solution turns milky B. a brick-red precipitate is formed C. there is no precipitate D. a blue- black colouration is produced.
72. Which of the following minerals contains fluorine as one of its constituent element A. cryolite B. Bauxite C. Kaolin D. Mica.
73. Which of the following pieces of apparatus can be used for drawing a current of air through a Equid? A. wash bottle B. Aspirator C. pipette D. capillary tube.
74. The green house effect is a climate condition associated with the pressure of excess A. carbon (IV) oxide B. hydrogen sulphide C. ammonia gas D. nitrogen (II) oxide.
75. Electrovalent compounds are characterized by A. solubility in ethanol B. high molar mass C. high melting point D. strong oxidizing ability.
76. A germinating seed requires oxygen which is essential for A. converting carbohydrate into glucose B. transporting energy from one part of the plant to another C. the production of energy by oxidizing essential carbohydrates D. hydrolysis of proteins
77. A carbohydrate molecules in the right vent ride of the heart is pumped into the cell of the toe of man. Which of these structures is it unlikely to pass through en route? A. pulmonary artery B. lungs C. Heart D. Liver.
78. Which of these elements is required by plants in small quantities only? A. Boron B. potassium C. Magnesium D. Nitrogen.
79. The mammalian stomach can carry out a number of possesses except: A. absorption of glucose B. secretion of hydrochloric acid C. production of trypsin D. churning of food.
80. A farmer working in a swamp rice farm did not have any food not drink any water. Which of the disease can he not contract? A. cholera B. bilharzias C. malaria D. river blindness.
81. A secchi disc is used in the determination of: A. Rainfall B. tidal wave C. turbidity D. current velocity.
82. Which of the these is not an enzyme? A. pepsin B. gastrin C. Amylase D. Trypsin.
83. The plumule is best described as: A. part of the cotyledon B. part of the endosperm C. a miniature shoot D. structure opposite the hilum.
84. A gland in the human body secretes two hormones, one of which causes a disease called diabetes when there is a deficiency. Which of these organs produces the hormones. A. spleen B. Adrenal body C. pancreas D. Thyroid and parathyroid.
85. A sugar solution was boiled with fehlings solutions A and B and the colour remained blue. The sugar tested was A. glucose B. maltose C. fructose D. Sucrose.
86. Root hairs have their origin from A. endodermis B. cortex C. xylem D. Epidermis.
87. The one seeded fruit in which the pericarp and seed have become fused together is known as A. achene B. samara C. caryopsis D. Drupe.
88. Milk can be pasteurized by A. the addition of sodium chloride B. boiling for one hour C. heating to 75°C for 20 minutes D. beating to 75°C for one hour.
89. Human beings require vitamins in their diet because vitamins A. contain carbohydrates and fats B. stimulate the alimentary canal C. digest proteins in the body D. influence many important chemical processes in the body.
90. The organ through which food and oxygen are supplied to the human embryo and which also prevents harmful materials from reaching the embryo is the A. Amnion B. Uterus C. placenta D. umbilical cord.
91. Which of the following food substance is digested in the stomach? A. carbohydrates B. fats and oils C. fats and proteins D. carbohydrates and fats.
92. The conversion of excess amino acids into urea occurs in the A. kidney B. pancreas C. spleen D. Liver.
93. Where is the food stored in a yam plant? A. stem B. Bud C. Adventious Root D. Tap root.

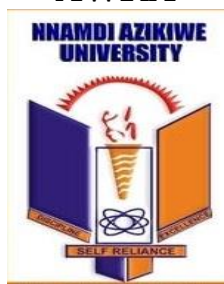
94. Growing yam tendrils climb for support. This growth response is A. haptotropism B. geotropism C. hydrotropism D. Chelliotropism.
 95. The functional unit of a mammalian kidney is the A. malphigian capsule B. glomerulus C. ureter D. Nephron.
 96. The glottis is the opening which leads to the A. oesophagus B. larynx C. mouth D. Pharynx.
 97. If the gall bladder of a man is removed by surgery, which of the following processes will be most seriously affected? A. Digestion of fats and oils B. Digestion of starch C. conversion of proteins D. formation of urea
 98. For pollen to be released in *Crotalaria iuncea* the insect must depress the A. wing B. keel C. standard D. antesealous stamen.
 99. In an agricultural ecosystem, the biotic component consists of. A. Crops, pest and beneficial insect B. Crops,
- | | | | | | | | |
|---|-------|-------|-------|-------|-------|-------|-------|
| A | 15. B | 16. A | 17. C | 18. D | 19. C | 20. C | 21. C |
| B | 30. C | 31. C | 32. D | 33. B | 34. D | 35. A | 36. C |
| D | 45. C | 46. C | 47. C | 48. D | 49. D | 50. B | 51. B |
| B | 60. C | 61. A | 62. C | 63. A | 64. A | 65. C | 66. C |
| A | 75. C | 76. C | 77. A | 78. A | 79. A | 80. A | 81. C |
| D | 90. C | 91. A | 92. D | 93. A | 94. B | 95. D | 96. D |

- temperature and humidity C. Pest, beneficial insect and water D. Crops, water and soil.
100. Which of the following is an example of discontinuous variation? A. skin coloration B. left-handedness C. Body weight D. Height

**ANSWERS TO MALI POST-LIME SCREENING
EXERCISE (2012/2013) SESSION
FOR AGRICULTURE, HEALTH SCIENCE AND
TECHNOLOGY, MEDICINE,
NATURAL SCIENCES AND PHARMACEUTICAL
SCIENCES**

- | | | | | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|-------|-------|-------|
| 1. D | 2. D | 3. A | 4. D | 5. D | 6. C | 7. D | 8. A | 9. C | 10. C | 11. D | 12. B | 13. D | 14. D |
| 22. C | 23. B | 24. A | 25. D | 26. C | 27. A | 28. B | 29. B | 30. C | 31. C | 32. D | 33. B | 34. D | 35. A |
| 36. C | 37. B | 38. D | 39. D | 40. D | 41. D | 42. D | 43. A | 44. D | 45. C | 46. C | 47. C | 48. D | 49. D |
| 50. B | 51. B | 52. C | 53. A | 54. C | 55. C | 56. B | 57. D | 58. B | 59. D | 60. C | 61. A | 62. C | 63. A |
| 64. A | 65. C | 66. C | 67. C | 68. D | 69. C | 70. D | 71. B | 72. A | 73. B | 74. D | 75. C | 76. C | 77. A |
| 78. A | 79. A | 80. A | 81. C | 82. C | 83. C | 84. C | 85. D | 86. D | 87. C | 88. C | 89. D | 90. C | 91. A |
| 92. D | 93. A | 94. B | 95. D | 96. D | 97. A | 98. B | 99. A | 100. B | | | | | |

**NNAMDI AZIKIWE UNIVERSITY,
AWKA**



POST-LIME SCREENING EXERCISE, 2013/2014.

FOR

**MEDICINE, NATURAL SCIENCES, HEALTH SCIENCES
AND TECHNOLOGY**

INSTRUCTIONS

1. Shade Firmly only one answer using HB pencil only
2. You are allowed to make a second choice of course into any of the following: Anatomy, Physiology, Pharmaceutical Sciences, Science Education, Natural Sciences (See the back of the OMR sheet for details)
3. Shade your choice of course in the column provided (column 8)

Time allowed: 1 hour

Date: July 24,2008

Choose the option opposite in meaning to the word(s) underlined

1. The government is making concreted efforts at providing essential social facilities in the rural areas. A. dissipated B. unconcerned C. Unsuccessful D. Uncontrolled,
2. The Western allies frowned at their enemies indiscriminate bombing of their territory A. impartial B. selective C. unprovoked D. Divided.
3. The girl is very intelligent, but her jokes are rather outrageous. A. offensive B. courageous C. pleasant D. Unwholesome.
4. We cannot but talk about his invaluable contributions to the affairs of the society.

- A. worthless B. Costly C. Unrecognized D. Incalculable.
5. This average fertility figure of course conceals wide individuals variations among the people. A. attracts B. covers C. exposes D. Concludes.
6. Men Living in an extended family know that have to be responsible for the welfare of some other person's children A. Extensive B. nuclear C. neutral D. Contracted.
7. The appearance of the pall-bearers was extremely fascinating. A. magnetic B. unorganized C. interesting D. Repulsive.
8. Many miners protested against the order to march into the coal pits A. strut B. amble C. trudge D. Ramble.
9. The nasty action will bring nothing but discredit to them A. honour B. shame C. reward D. Disgrace.
10. The man's health has deteriorated in the hospital A. collapsed B. improved C. worsened D. Revived.
11. The boss made a clean sweep of all the old hands in the office.1 A. completely got rid B. clearly swept the office C. made a quick change D. made an adjustment.
12. Her anxiety was apparent to every one present at the ceremony. A. real B. visible C. transparent D. Unknown.
13. Our experience on board the aircraft is nothing to write home about. A. not remarkable B. not unpleasant C. nobody's business D. not a home affair.
14. The rampage was organized and directed by the militant ones among the students. A. soldiers B. Rebellious ones C. belligerent ones D. Commandoes.
15. The governor advised his people to get off the beaten tracks in their approach to community matters. A. use the old road B. make the tracks better C. find new ways D. avoid beating about the bush.
16. It was their custom to keep an open house on Sundays. A. welcome all visitors B. keep the shutters open C. leave the house open D. expose the whole house.
17. The president made a passionate appeal for calm. A. in flamed B. disturbing C. intense D. Lively.
18. Dauda stole the clay taking from the bakery. A. receipts B. collections C. contributions D. Offerings.
19. You do not need to stutter in order to show your excitement. A. jump about B. shout C. sob D. Stammer.

20. What embarrasses me is the stealthy way in which the new servant goes about his duties in this house. A. indolent B. furtive C. boisterous D. Agile.
21. The way he handled the whole issue was rather despicable. A. disturbing B. likeable C. contemptible D. Frustrating.
22. The authenticity of the reporters claims should be established before taking further actions on the matter. A. facts B. source C. evidence D. Genuineness.
23. After careful investigation, the police found no iota of truth in Ibrahim's allegation A. quota B. grain C. evidence D. Statement.
24. The case was thrown out because the court lacked jurisdiction. A. jurors to help the judge B. authority C. prosecutors D. appellate powers.
25. The chief will launch the fund-raising appeal. A. make a speech at B. eat his afternoon meal during C. travel by boat to D. start off.
26. Okeke has been employed but he is unhappy because it is a run of mill job. A. boring B. a poor paying C. a tiring D. an ordinary.
27. I would like to believe that this brilliant performance of yours is not just a flash in the pan. A. repeated occurrence B. normal happening C. flashy happening D. chance occurrence.
28. As the wedding day approached the bride began to develop cold feet. A. fall sick B. feel cold C. be reluctant D. become aggressive.
29. I told my brother to get the matter of his chest as soon as possible. A. give serious thought to the matter B. say what has been anxious to say C. make a brief comment D. state his points clearly.
30. He has a big heart but he is inept at following a witty conversation. A. large B. huge C. warm D. Fragile.
31. The clerk refused to answer for the mistakes made by the manager and his assistants. A. reply to B. give an answer C. accept responsibility for D. account for.
32. Most of his observations were wide of the mark. A. comprehensive B. irrelevant C. pertinent D. Unacceptable.
33. I didn't think she could be so easily taken in by his pretenses. A. flattered B. impregnated C. deceived D. Overcome.
34. Do you have the same aversion as I do for war films? A. bitterness B. dislike C. criticism D. Preference.
35. His summary of the meeting was brief and to the point. A. precise B. accurate C. exact D. Crucial.
36. The Minister hit on a plan to retain his post after many months of lobbying. A. drew up a plan B. discovered a plan C. selected a plan D. designed a plan.
37. My friend will hate his uncle forever because he left him in the lurch in his hour of need. A. abandoned him B. punished him C. disinherited him D. despised him.
38. Our school prefect is too officious and we all hate him because of his behaviour. A. efficient B. efficacious C. over-zealous D. Active

Choose the word(s) or phrase(s) which best fill(s) the gap(s)

39. He regarded the betrayed of his friend as a stab in the back. A. on B. at C. in D. From.
40. Our manager has instructed that all customers must be barred from our bank. A. those type B. that type C. that type s D. this types

41. He was not a very good student when I first met him but the year has ultimately changed his character. A. mature/mature/mature B. mature/matured C. mature/matured D. Matured/matured.
42. Because of the _____ he did to my car, I am going to sue him for _____.
A. damage/damages B. damages/damage C. damages/damages D. Damage/damage.
43. Although I am watching television, I _____ what you are saying. A. am hearing B. can hear C. have heard D. was hearing.
44. One of the rioters who _____ arrested yesterday _____ released. A. was/had been B. were/have been C. was/have been D. were/has been.
45. The teacher invited Ifeanyi and _____ to his office. A. myself B. me C. I D. Himself.
46. Joseph and his wife are always fighting. The _____ to be drawn from this is that they are not happy together. A. reference B. inference C. difference D. Deference.
47. The government has provided new _____ for the local government polls. A. guard lines B. guide lines C. guard lines D. Guide lines.
48. The manager was angry with Okoye for _____ seeds of discord among the workers. A. planting B. sewing C. cultivating D. Sowing.
49. I looked for you all over the campus but I could not find you. A. find B. See C. meet D. Catch.
50. _____ is always superior to any other in town. A. there's B. Theirs C. their's D. There.
51. Two radioactive elements A and B have half-lives of 100 and 50 years respectively. Samples of A & B initially contain equal number of atoms. What is the ratio of the number of the remaining atoms of A to that of B after 200 years? A. 4:1 B. 2:1 C. 1:1 D. 1:4
52. A boy standing some distance from the foot of a tall cliff claps his hands and hears an echo 0.5s later. If the speed of sound is 340ms⁻¹, how far is he from the cliff? A. 17m B. 34m C. 85m D. 170m.
53. Which of the following is not correct about isotopes of an element? They have A.. the same neutron number B. the same proton number C. the same number of electrons D. different physical properties.
54. Which of the following is not electromagnetic radiation? A. X-ray B. Radio waves C. sunlight D. sound waves.
55. Cloud formation is the direct result of A. precipitation B. vaporization C. condensation D. Sublimation.
56. Water boils at a lower temperature when heated at the top of a mountain than at sea-level because at the top of the mountain the A. saturation vapour pressure of water is higher than at sea-level B. sun adds more heat to the water C. relative humidity of the atmosphere is lower D. pressure of the atmosphere is lower.
57. The main reason why rice cooks faster in a pressure cooker than in a cooking pot is that A. less heat escapes from the cooker B. the vapour pressure in the cooker is constant C. the boiling point of water in the cooker is raised D. the vapour pressure in the cooker decreases.
58. In which of the following media would sound waves travel faster? A. kerosene B. alcohol C. iron D. Air.
59. Which of the following phenomena is the practical evidence for the existence of the continual motion of molecules? A. Translational motion B. Rotational motion C. Brownian motion D. Oscillatory motion.
60. Each of the following physical quantities is classified as a vector or scalar quantity. Which of the classifications is

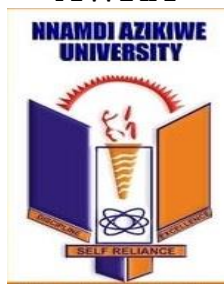
- connect? A. Electric potential (vector) B. Momentum (scalar) C. Gravitational field intensity (scalar) D. magnetic flux density (vector)
61. The friction which exists between two layers of liquid in relative motion is called A. capillarity B. surface tension C. viscosity D. Cohesion.
62. Which of the following simple machines is a second class lever? Wheel barrow B. crowbar C. claw hammer D. Seesaw.
63. Which of the following types of motion does a body 'undergo when acted upon by a couple? A. vibrational B. translational C. Rotational D. Random.
64. The frequency of a swinging pendulum is the A. number of complete oscillations the pendulum makes in one second B. number of amplitudes the bob makes in on second C. angle the bob swings through in one second D.. distance the bob covers in one second.
65. Which of the following methods quantities has the same units as energy? A. power B. work C. force D. Momentum.
66. Which of the following methods can be used to separate a mixture of two miscible liquids with different boiling points? A. Decantation B. Distillation C. evaporation D. Filtration.
67. Pig-iron is brittle because it contains A. a high percentage of carbon as impurity
B. calcium trioxosilicate (iv) C. unreacted haematite D. Undecomposed limestone.
68. Which of these metals will not liberate hydrogen from dilute HCl ? A. copper B. Iron C. Magnesium D. Zinc.
69. Radon is used as a tracer in medical research because it A. is monatomic B. is radioactive C. is a noble gas D. has a complete valence shell.
70. In linear molecules, the bond angle is A. 90° B. 104° C. 180° D. 20° .
71. A sodium atom and a sodium ion have the same A. number of neutrons B. number of electrons C. Electric charge D. chemical properties.
72. An increase in the pressure of a gas results in a decrease in it's A. mass B. vapour density C. volume D. Temperature.
73. An acid is a substance which in the presence of water produces A. salts B. oxygen C. hydroxonium ions D. effervescence.
74. Which of the following is a general method of preparing acid? A. Direct combination of constituent elements B. Double decomposition involving a salt solution C. Reaction between an anhydride and water D. Dissolution of hydroxides followed by neutralization.
75. Which of the following ions will migrate to the cathode during electrolysis? A. Zinc ion B. Chloride ion C. sulphide ion D. tetraoxosulphate (vi) ions.
76. What quantity of electrons (in mole) is lost when one mole of iron(II) ions oxidized to iron (III) ions? A. 5 moles B. 4 moles C. 1 mole D. 3 moles.
77. The common feature of reactions at the anode is that A. electrons are consumed B. oxidation is involved C. ions are reduced D. the electrode dissolves.
78. Ethane undergoes mainly addition reactions because it is A. a gas B. a hydrocarbon C. unsaturated D. a covalent compound.
79. The position of equilibrium in a reversible reaction is affected A. particle size of the reactants B. change in concentration of the reactants C. presence of a catalyst D. change in size of the reaction vessel.
80. The reactions between alkanolic acids and alkanols in the presence of a mineral acid is known as A. saponification B. Hydrolysis C. Polymerization D. etherification.
81. A fundamental similarity between nervous and hormonal system is that both A. involve chemical transmission B. have widespread effects C. shed chemical into the blood stream D. evoke rapid response
82. The irreversible life process by which new protoplasm is added to increase the size and weight of an organism can be termed A. anabolism B. catabolism C. growth D. Development.
83. If an organism obtain its food by means of haustoria, it is to be A. holophytic B. heterophytic C. saparophytic D. Parasitic.
84. The main waste products formed in plants are A. alkaloids tannins and resins B. water, alkaloids and carbon dioxide C. oxygen, carbon dioxide and alkaloids D. water, carbon dioxide and oxygen.
85. In the mammalian respiratory system, exchange of gases occurs in the A. Lungs B. bronchi C. Bronchioles D. Alveoli.
86. If the phloem of a healthy plant is killed by heat treatment, the A. upward movement of salts will cease B. food manufacture in the leaves will accumulate C. whole plant will die immediately D. leaves of the plant will become yellow.
87. If an animal has diastema, it would not possess A. incisors B. Canines C. Premolars D. Molars.
88. The major function of the swim bladder in fishes is A. breathing B. buoyancy C. swimming D. Diving.
89. Insects and millipedes have many features in common except A. jointed appendages B. compound eyes C. segmental body D. Exoskeleton.
90. What is the functions of trichocyst in paramecium A. movement B. defense C. excretion D. Reproduction.
91. The sum of the angle of a regular polygon is 2520° . how many sides does the polygon have? A. 20 B. 18 C. 15 D. 16.
92. If the exterior angles of a quadrilateral are, $(2y+5)^\circ$, $(y+15)^\circ$ and $(3y-10)^\circ$, find Y A. 61.43 B. 60 C. 51.43 D 50.
93. Calculate the surface area of a sphere of radius 7cm (take $\pi = \frac{22}{7}$). A. 86cm^2 B. 154cm^2 C. 616cm^2 D. 143cm^2 .
94. If $3 \log a + 5 \log a - 6 \log a = \log 64$, what is a? A. 4 B. 6 C. 8 D. 16.
95. A sales girl gave a change of N1.15 to a customer instead of N1.25. Calculate her percentage error A. 10% B. 2.4% C. 8% D. 0.8%.
96. The ages of three men are in the ratio 3:4:5. If the difference between the ages of the oldest and youngest is 18 years, find the sum of the ages of the three men. A. 45 years B. 72 years C. 108 years D. 216 years.
97. Which of the following numbers is a perfect cube. A. 350 B. 504 C. 950 D. 1728 A car moves at an average speed of 30km/h how long does it take to cover 200 metres? A. 2.4 sec B. 24 sec C. 144 sec D. 240 sec.
98. If N varies directly as M and N = 8 when M = 20, find M when N = 7 A. 13 B. 15 C. $17\frac{1}{2}$ D. 18.
99. Express 0.0462 in standard form. A. 0.462×10 B. 0.462×10^{-1} C. 4.462×10^{-1} D. 4.62×10^2

ANSWERS TO NAU POST-LIME SCREENING EXERCISE (2013/2014)

**SESSION FOR MEDICINE, NATURAL SCIENCES, HEALTH SCIENCES
AND TECHNOLOGY**

1. A	2. B	3. D	4. A	5. C	6. B	7. D	8. C	9. A	10. B	11. A	12. B
13.A	14.C	15.C	16.A	17.C	18.A	19.D	20.B	21.C	22.D	23.C	24.D
25.D	26.D	27.D	28.C	29.B	30.A	31.C	32.B	33.C	34.B	35.A	36.B
37.A	38.C	39. C	40. A	41. A	42. A	43. B	44. D	45. B	46. B	47. D	48. D
49. B	50. B	51. B	52. C	53. A	54. D	55. C	56. A	57. A	58.C	59.C	60.C
61.C	62.A	63.C	64.A	65.B	66. B	67. A	68. A	69. B	70. C	71. A	72. C
73. C	74. C	75. A	76. D	77. B	78. C	79. B	80. B	81. C	82. C	83. D	84. A
85. D	86. B	87. B	88.C	89. B	90. B	91. D	92. D	93. C	94. C	95. C	96. C
97. D	98. B	99. C	100. D								

**NNAMDI AZIKIWE UNIVERSITY,
AWKA**



**POST-UME SCREENING EXERCISE, 2014/2015
FOR**

**MEDICAL, NATURAL SCIENCES, HEALTH SCIENCES
AND TECHNOLOGY,
BASIC MEDICINE SCIENCES AND PHARMACEUTICAL
SCIENCES**

INSTRUCTIONS

1. Shade (FIRMLY) only one answer using NB pencil only
 2. You are allowed to make a second choice of course into any of the following: Anatomy, Physiology, Pharmaceutical Sciences, Science Education (see the back of the OMR sheet for details).
 3. Shade your second choice of course in the column provided (column 8)
- Time allowed: 1 hours

Date: July 7, 2009

Choose the option opposite in meaning to the word underlined in each of the following sentences

1. Okoro is an amateur wrestler (A) skillful (B) good (C) professional (D) strong.
2. The instructions on the examination paper are explicit. (A) simple (B) ambiguous (C) detailed (D) definite.
3. The doctor tried to alleviate his patient's pain (A) relieve (B) worsen (C) kill (D) cure.
4. Ojo's response infuriated his wife. (A) annoyed (B) pleased (C) surprised (D) confused.
5. He accepted a mundane task without hesitation. (A) great (B) lowly (C) menial (D) moderate.
6. It is a unique opportunity for her to demonstrate the reality of her faith (A) strange (B) usual (C) golden (D) unusual.
7. Her identification with the king is publicly known. (A) hatred (B) disassociation (C) relationship (D) intimacy
8. The economic situation in Nigeria is obviously gloomy. (A) encouraging (B) moody (C) unknown (D) regrettable.
9. One wonders if the situation will improve. (A) brighten (B) diminish (C) disintegrate (D) worsen.
10. Language teachers believe that grammar exercise stretch the mind. (A) expand (B) ruin (C) enrich (D) restrict.
11. These two books are identical (A) alike (B) similar (C) different (D) equal

Choose the option nearest in meaning to the word(s) or phrase(s) underlined in the following sentences

12. The police ran the criminal on earth (A) jailed him (B) knocked him down (C) discovered him (D) buried him.
13. If your life is a turmoil, always take courage. (A) devastation (B) crisis (C) trail (D) tragedy.
14. Do you know one of the most astounding events of my life? (A) special (B) amazing (C) serious (D) outstanding.
15. Nneka is suffering from the consequences of alienation (A) confinement (B) isolation (C) enclosure (D) imprisonment.

16. The terms of the contract struck in my throat (A) were beyond me (B) were not clear (C) were ambiguous (D) became obsolete.
17. He was asked to give copies examples to appear convincing (A) concrete (B) rigid (C) cogent (D) many.
18. Some children mimic their teachers. (A) imitate (B) mime (C) ridicule (D) tease.
19. He works long hours collecting trash to eke out a livelihood. (A) try a living (B) struggle for a living (C) make a living (D) carve out a living.
20. Everybody complained of a lean harvest last year. (A) surplus (B) abundant (C) poor (D) thin.
21. He is modest in his demands (A) honest (B) bogus (C) extravagant (D) humble.
22. The patient disregarded the advice of the doctor. (A) ignored (B) disobeyed (C) questioned (D) respected.
23. The newly elected leader has pledged to ensure better life for the citizens. (A) vowed (B) agreed (C) undertaken (D) undertaking.
24. Statement are revered for their objectivity. (A) referred (B) respected (C) remembered (D) rejected.
25. He has advised to keep his head the confusion notwithstanding. (A) avoid being beaten or insulted (B) keep calm (C) save 1head (D) prevent being beheaded.
26. Tuned always puts his shoulder to the wheel. (A) sits with his shoulder straight when he is driving (B) works energetically at the takes in hand (C) performs task assigned to him grudgingly (D) holds the steering wheel firmly when he is driving.
27. He reneged on the agreement between him and his employees. (A) kept (B) failed to keep (C) failed to approved (D) failed to sign.
28. He is credulous. (A) credible (B) creditable (C) gullible (D) fallible.
29. The company has gone under. (A) suffered some losses (B) broken up (C) becomes broke (D) become bankrupt.

Fill the gap with appropriate option in each of the following sentences

30. Many people believe that nuclear power will solve our energy problems. This has not been proved to be true. (A) Indeed (B) But (C) However (D) On the contrary.
31. Sign post words are useful to renders. (A) Conclusively (B) To conclude (C) In conclusion (D) The conclusion.
32. Every programming language and software package _____ limitations (A) have its (B) have their (C) has its (D) has their.
33. A programme of good exercise may help a person fight _____ cold. (A) out (B) at (C) with (D) off.
34. Baba and _____ participated in the tournament. (A) him (B) his (C) he (D) he's.
35. It was _____ who fought the civil war. (A) they (B) them (C) those (D) theirs.
36. The physicians have more people in _____ waiting rooms than _____ have ever had. (A) his/he (B) there, they (C) their/they (D) them/they.
37. Before mechanization, workers _____ water out of fabrics with their hands. (A) wring (B) wrung (C) wrang (D) wringed.
38. Four engineers _____ on this system since March. (A) worked (B) are working (C) had worked (D) have been working.

39. Having worked all night, the security man _____ a sense of accomplishment. (A) had felt (B) felt (C) is feeling (D) has felt.
40. Three quarters of the Physics class _____ dramatically. (A) improve (B) improves (C) are improving (D) is improving.
41. A number of student; _____ missed the opportunity to re-register carryover course. (A) is (B) has (C) have (D) do.
42. It seems to be a well thought _____ scheme (A) over (B) out (C) off (D) into.
43. Edward is prepared for a _____ with his opponent following his defeat last season. (A) show-off (B) show-down (C) show-out (D) show-up.
44. Victor refuse to be _____ though he has written the same examination three times. (A) sad (B) placated (C) frustrated (D) indifferent.
45. On his _____ of office, the new President announced some drastic measures. (A) assumption (B) ascension (C) acceptance (D) appointment.
46. The _____ was filed _____ the Enugu Magistrate Court. (A) suit/in (B) case/in (C) suit/at (D) case/of.
47. The plaintiff _____ the court to restrain the defendants from further action. (A) asked (B) begged (C) demanded of (D) prayed.
48. My experience in Lagos last week was _____ about. (A) something to (B) nothing to explain at home (C) nothing to write home (D) something to celebrate about.
49. _____ Him around, I would have informed you. (A) Had I seen (B) Have I seen (C) Should see (D) If I saw.
50. Adam is one of the boys who always _____ ghost work. (A) does (B) would do (C) do (D) done

Identify the correct answer to each question below:

51. The density of 400cm³ of palm oil was 0.9g/cm³ before frying. If the density of the oil was 0.6g/cm³ after frying. Assuming no loss of oil due to spilling, its volume was (A) 360cm³ (B) 600cm³ (C) 240cm³ (D) 800cm³.
52. If a given mass of gas at constant pressure obeys the relation — volume is proportional to the absolute temperature — the gas is said to obey (A) Hook's law (B) Boyle's law (C) Newton's law (D) Charles's law.
53. Two closed vessels contain equal volume of boiling water and are identical except that one is blackened on the outside while the other is polished. Which of the following is correct? (A) The polished vessel will cool faster because a polished surface is a good radiator (B) The polished vessels will cool faster because it will absorb less heat from the room (C) The black vessel will cool faster because it is a better conductor of heat (D) The polished vessel will cool faster because it reflects heat better.
54. When the vapour of a substance is in equilibrium with its own liquid, it is said to be (A) gaseous (B) diffused (C) saturated (D) unsaturated.
55. Which of the following statements is correct? The mgnestized state of an iron bar can be destroyed completely by (A) heating the bar to a high temperature (B) breaking the bar (C) connecting opposite poles with a small piece of iron (D) placing the inside a strong magneistic field.
56. An electric cell has an internal resistance of 2ohms. A current of 0.5 A is found to flow when a resistor of 5ohms resistance is connected across it. What is the electromotive

- force of the cell? (A) 5 volts (B) 3.5 volts (C) 2.5 volts (D) 1.0 volt.
57. Which of the following is the correct definition of "half-life"? (A) The time it takes a radioactive material to decay to half of its original quantity (B) the average life-time of a radioactive material (C) Half the period a radioactive material decays completely (D) Half of the decay constant.
58. A concave lens of focal length 20cm forms an image $\frac{1}{2}$ the size of the object. The object distance is (I) 100cm (B) 100/9cm (C) 60cm (D) 60/7cm.
59. Which of the following is true of atoms? (A) Atoms are indestructible (B) All atoms of a pure substance or element are identical (C) Atoms of difference substances have different weights (D) All of the above.
60. Which of the following statement is true of the law of universal gravitation? (A) Any two bodies attract each other with a force which is proportional to the difference of their masses (B) Any two bodies attract each other with a force which is inversely proportional to the square of the distance between them (C) Any two bodies attract each other with a force which is proportional to the product of their masses and inversely proportional to the square of the distance between them. (D) Any two bodies attract each other with a force which is directly proportional to the square of the distance between them.
61. Zinc oxide is a (A) basic oxide (B) acidic oxide (C) amphoteric oxide (D) neutral oxide.
62. When sodium chloride and metallic sodium are dissolved in water (A) both processes are exothermic (B) both processes are endothermic (C) the dissolution of metallic sodium is exothermic (D) the dissolution of metallic sodium is endothermic.
63. The oxidation state of chlorine in potassium chlorate is (A) +a (B) +2 (C) -3 (D) +5.
64. When air, which contains the gases oxygen, nitrogen, carbon dioxide, water vapour and the rare gases, is passed through alkaline pyrogallol and then over quicklime, the only gases left are (A) nitrogen and carbon dioxide (B) nitrogen and the rare gases (C) the rare gases 1(D) nitrogen and oxygen.
65. When heat is absorbed during a chemical reaction, the reaction is said to be (A) thermodynamic (B) exothermic (C) isothermic (D) endothermic.
66. The nucleus of an atom contains (A) protons only (B) neutrons only (C) protons and electrons (D) protons and neutrons.
67. Hydrochlorous acid is used as a bleach because (A) it is a strong acid (B) it yields chlorine readily in pure water (C) it is an oxidizing agent (D) it is a reducing agent.
68. The most common type of chemical reaction which alkanes undergo is (A) substitution (B) addition (C) condensation (D) polymerization.
69. In titration involving sodium hydroxide solution and dilute hydrochloric acid. Where would you place the base? (A) in the beaker (B) in the conical flask (C) in the burette (D) in the standard flask.
70. Which of the following substances are all made by the process of polymerization? (A) Nylon and soap (B) ethanoic acid, margarine and ethanol (C) Nylon and artificial rubber (D) Nylon and margarine.
71. Plants adapted for life in salty marsh are called (A) hydrophytes (B) Xerophytes (C) halophytes (D) epiphytes.

- ch of these is the terminal portion of the alimentary canal of a mammal? (A) Esophagus (B) Stomach (C) Rectum (D) Appendix.
73. All living organisms (A) photosynthesize (B) respire (C) move (D) feed.
74. Which of these tissues serves the function of support and water conduction? (A) Parenchyma (B) Collenchyma (C) Xylem (D) Sclerenchyma.
75. Which of the substances listed below is a trace element from plants? (A) Potassium (B) Sodium (C) Copper (D) Nitrite.
76. A Secchi disc is used in the determination of (A) tides (B) waves (C) current velocity (D) turbidity.
77. The carrier of the hereditary characters in plants and animals is the (A) cell (B) nucleus (C) chromosome (D) gene.
78. Which of these is not an enzyme? (A) Pepsin (B) Gastrin (C) Amylase (D) Chymotrypsin.
79. Which of these is not an excretory organ? (A) Lungs (B) Kidney (C) Leaf (D) Large intestine.
80. Which of these disease cannot be controlled killing the sector (A) River blindness (B) Malaria (C) Bilharzia (D) Cholera.
81. A sugar solution was boiled with Fehling's solution A and B and the colour remained blue. The sugar tested was (A) glucose (B) maltose (C) fructose (D) sucrose.
82. Alkaline pyrogallol was used in an experiment. That experiment must have been connected with (A) excretion (B) germination (C) digestion (D) photosynthesis.
83. The plumule is best described as (A) part of the cotyledon (B) part of the endosperm (C) a miniature shoot (D) structure near the micropyle.
84. A dry indehiscent winged fruit formed from one carpel is known as (A) schizocarp (B) caryopsis (C) samara (D) nut.
85. A gland in the human body secretes two hormones, one of which causes a disease called diabetes when there is deficiency. Which of these organs produces the hormone? (A) spleen (B) thyroid (C) pancreas (D) adrenal.
86. One of the following statements is not true of viruses. (A) they are microorganisms (B) they are smaller than bacteria (C) they can be seen with an ordinary light microscope (D) they cause polio and small pox.
87. The digestive enzyme that coagulates proteins in milk is (A) rennin (B) ptyalin (C) pepsin (D) trypsin.

88. Osmic acid boiled with a solution of food substance gave a black precipitate. This indicates the presence of (A) proteins (B) fats and oils (C) amino acids (D) starch.
89. Members of the phylum protozoa use the contractile vacuole to (A) remove excess food (B) movement (C) remove excess water (D) for digestion.
90. The main function of the petal of a flower is to (A) attract pollination agents (B) protect the flower while still in bud (C) serve as landing stage for insects (D) protect inner parts from desiccation.
91. In one and half hours, the minute hand of a clock rotates through an angle of (A) 900 (B) 180° (C) 5400 (D) 6400.
92. A group of 14 children received the following scores in a reading test; 35, 3526, 26, 29, 29, 12, 25, 25, 25, 25, 17. What was the median score? (A) 29 (B) 26 (C) 23 (D) 25.
93. By selling an article for N45.00 a man makes a profit of 8%. For how much should he have sold it in order to make a profit of 32%? (A) N 180.00 (B) 59.00 (C) N55.00 (D) 58.00.
94. Seven (7) pupils of average age 12 years leave a class of 25 pupils of average age of the pupils now in the class? (A) 13 years (B) 13 years 5 months (C) 13 years (D) 11 years.
95. A school girl spends $\frac{1}{4}$ of her pocket money on books and $\frac{1}{3}$ on dresses. What fraction remains? (A) $\frac{5}{6}$ (B) $\frac{5}{12}$ (C) $\frac{7}{12}$ (D) $\frac{1}{6}$.
96. Each interior angle of a rectangular polygon is 108°. How many sides has it? (A) 5 (B) 7 (C) 9 (D) 14.
97. Find correct to two decimal places, the mean of 9, 13, 16, 17, 23, 24 (A) 23.00 (B) 17.29 (C) 16.50 (D) 16.33.
98. A boy walked 800m in 20 minutes. Calculate his average speed in km per hour. (A) 2.4km per hour (B) 4.0km per hour (C) 16km per hour (D) 24km per hour.
99. The sides of two cubes are in the ratio of 5:5. What is the ratio of their volumes? (A) 4:5 (B) 8:15 (C) 6:125 (D) 8:125.
100. A man bought 220 mangoes at N5x. He sold each for 3x kobo and made a gain of N8.00. Find the value of x. (A) 2 (B) 5 (C) 6 (D) 10

ANSWERS TO NAU POST-LIME SCREENING EXERCISE (2014/2015)
SESSION FOR MEDICINE, NATURAL SCIENCES, HEALTH SCIENCES AND
TECHNOLOGY, BASIC MEDICINE SCIENCES AND PHARMACEUTICAL
SCIENCES

1. C	2. A	3. B	4. 8	5. A	6. B	7. B	8. A	9. D	10. B	11. C	12. A
13. B	14. A	15. B	16. B	17. B	18. A	19. C	20. C	21. D	22. A	23. C	24. B
25. B	26. B	27. B	28. A	29. D	30. D	31. C	32. C	33. A	34. C	35. A	36. C
37. A	38. C	39. B	40. A	41. B	42. B	43. B	44. C	45. A	46. A	47. D	48. C
49. A	50. C	51. B	52. D	53. A	54. C	55. D	56. A	57. A	58. B	59. D	60. C
61. C	62. C	63. B	64. B	65. D	66. D	67. C	68. A	69. B	70. A	71. C	72. C
73. B	74. C	75. C	76. D	77. D	78. B	79. D	80.	81.	82.	83.	84.
85.	86. C	87. A	88. A	89. A	90. A	91.	92.	93.	94.	95.	96.
97.	98.	99.	100.								

NNAMDI AZIKIWE UNIVERSITY,
AWKA



PORT-LIME SCREENING EXERCISE, 2015/2016
FOR
MEDICINE, NATURAL SCIENCES, HEALTH SCIENCE
AND TECHNOLOGY,
BASIC MEDICAL SCIENCES AND PHARMACEUTICAL
SCIENCES

INSTRUCTIONS

Time Allowed: 1 hour 2010

Date: July 21,

1. Shade FIRMLY only one answer using HB pencil only
2. Shade your second choice of course in the column provided (Column 8)

From the list of words lettered A to D, choose the word or group of words that is most nearly opposite in meaning to the underlined word.

1. There is no doubt that pride is less desirable a quality than _____. (A) arrogance (B) honesty (C) efficiency (D) humility.
2. Few people prefer the rural to the _____ areas. (A) modern (B) urban (C) pastoral (D) rustic.:
3. Dan is quite refined in his manners but his colleague is rather _____. (A) arrogant (B) crude (C) Cultured (D) Unskillful.
4. Instead of helping to preserve our culture, the youth engage in activities that _____ it. (A) denounce (B) alter (C) conserve (D) destroy.
5. We should regard donations as _____ and not compulsory. (A) mandatory (B) necessary (C) voluntary (D) obligatory.
6. The affluence of the politicians contrasts with the _____ of the masses. (A) poverty (B) paucity (C) misery (D) suffering.
7. The importation of rice is permitted while that of wheat is _____. (A) allowed (B) suspended (C) inhibited (D) prohibited.
8. The goal scored in the second half of the match lifted the team from the agony of defeat to the _____ of victory. (A) hope (B) celebration (C) ceremony (D) ecstasy.
9. One species of the animal has gone into extinction while the other is still in _____. (A) antiquity (B) oblivion (C) existence (D) irrelevance.
10. The council ratified the controversial recommendation which we thought would be _____. (A) refuted (B) rejected (C) authenticated (D) corroborated.

FROM THE WORDS LETTERED A TO D, CHOOSE THE WORD THAT BEST COMPLETES EACH OF THE FOLLOWING SENTENCES.

11. He was sent to the Remand Home in the hope that the place would _____ him. (A) amend (B) improve (C) reform (D) remand.
12. Since there were no further comments, the Chairman called for the _____ of the minutes of the last meeting. (A) objection (B) correction (C) resolution (D) adoption.
13. There was a ten-minute _____ during the stage performance of the play. (A) interim (B) interplay (C) interlude (D) intercession.
14. The judge restrained the landlord from _____ the tenant. (A) ejecting (B) sacking (C) dismissing (D) attacking.
15. The soldiers pledged their _____ loyalty to their commander. (A) divide (B) unalloyed (C) remarkable (D) enormous.

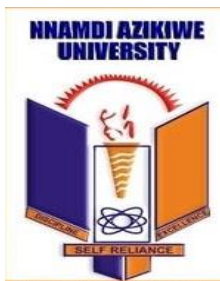
16. Many children are suffering from _____ because of poor feeding. (A) meningitis (B) maltreatment (C) hepatitis (D) malnutrition.
17. The elders advised the Priest to confine himself to his _____ duties. (A) sacrilegious (B) pastoral (C) angelic (D) divine.
18. The audience gave the actor a standing _____ for his beautiful performance. (A) proclamation (B) ovation (C) clapping (D) reward.
19. If the customer fails to honour his pledge again, his _____ will surely be in doubt. (A) responsibility (B) validity (C) credulity (D) credibility.
20. The judge warned that he would not tolerate any interruption of the _____ of the court. (A) proceeds (B) deliberations (C) proceedings (D) arguments.
21. Which of the following physical process cannot be explained by the molecular theory of matter? (A) Evaporation (B) Thermal conduction (C) Radiation of heat (D) Convectional currents in fluids.
22. A spring balance, which is suspended from the roof of a lift, carries a mass 1kg at its free end. If the lift accelerated upwards at 2.5ms^{-2} , determine the reading on the spring balance ($g = 10\text{ms}^{-2}$). (A) 25.0N (B) 12.5N (C) 7.5N (D) 4.0N.
23. An object of weight 10N immersed in a liquid displaces a quantity of the liquid. If the liquid displaced weights 6N, determine the up thrust on the object. (A) 20N (B) 10N (C) 6N (D) 4N.
24. A block weighing 15N rests on a flat surface and a horizontal force of 3N is exerted on it. Determine the frictional force on the block. (A) 0.3N (B) 0.5N (C) 3.0N (D) 5.0N.
25. The apparent weight of a body fully immersed in water is 32N and its weight in air is 96N. Calculate the volume of the body. (Density of water = 1000gm^3 , $g = 10\text{ms}^{-2}$). (A) $8.9 \times 10^{-3}\text{m}^3$ (B) $6.4 \times 10^{-3}\text{m}^3$ (C) $3.2 \times 10^{-3}\text{m}^3$ (D) $3.0 \times 10^{-3}\text{m}^3$.
26. The resistance of a wire depends on (A) the length of the wire (B) the diameter of the wire (C) the temperature of the wire (D) all of the above.
27. Which of the following is a correct explanation of the inertia of a body? (A) Reluctance to stop moving (B) Ability to overcome the earth's gravity (C) Reluctance to start moving and its reluctance to stop moving once it has begun to move (D) Readiness to start moving.
28. A device that converts sound energy into electrical energy is (A) the horn of a motor car (B) an A.C. generator (C) a microphone (D) a loudspeaker.
29. What is the cost of running five 50w lamps and four 100w lamps for 10 hours if electrical energy costs 2 kobo per KWh? (A) N0.65 (B) N0.13 (C) N3.90 (D) N1.30.
30. Which of the following is NOT a fundamental SI unit? (A) Meter (B) Kelvin (C) Ampere (D) radian.
31. A simple pendulum with a period of 2.0s has its length doubled. Its new period is (A) 1.00s (B) 28.3s (C) 2.83s (D) 0.35s.
32. Which of the following phenomena cannot be explained by the molecular theory of matter? (A) Expansion (B) Conduction (C) Radiation (D) Evaporation.
33. In a gas experiment, the pressure of the gas is plotted against the reciprocal of the volume of the gas at a constant temperature. The unit of the slope of the resulting curve is (A) force (B) force/m (C) work (D) energy/ m^2 .
34. Water shows anomalous behavior (A) below 0°C (B) between 0°C and 4°C (C) at exactly 4°C (D) above 100°C .

35. A heavy object is suspended from a string and lowered into water so that it is completely submerged. The object appears lighter because (A) the density of water is less than that of the object (B) the pressure is low just below the water surface (C) it experiences an up thrust (D) the tension in the string neutralizes an up thrust (D) the tension in the string neutralizes part of the weight.
36. Which of the following is a derived unit? (A) kilogram (B) Metre (C) Kelvin (D) Newton
37. A dentist obtains a linear magnification of 4 of a hole in a tooth by placing a concave mirror at a distance of 2.0cm from the tooth. The radius of curvature of the mirror is (A) 5.3cm (B) 3.2cm (C) 2.6cm (D) 1.5cm.
38. What is the effect of the increase in the size of the hole of a pin-hole camera on the image? It (A) gives a blurred image (B) corrects for chromatic aberration (C) magnifies the image (D) brings the image into sharper focus.
39. The mechanical advantages of an inclined plane depends on (A) its length (B) its height (C) the ratio of its length to its height (D) the product of its length and height.
40. A man walks 8km north and then 5km in a direction 60° east of north. Find the distance from his starting point. (A) 11.36km (B) 13.00km (C) 12.36km (D) 14.36km.
41. Glucose reduces Fehling's solution on warming to (A) copper (I) oxide (B) copper (II) oxide (C) copper (I) chloride (D) copper (II) hydroxide.
42. Pig-iron is brittle because it contains (A) a high percentage of carbon as impurity (B) calcium trioxosulphate (VI) acid (C) lead (IV) tetraethyl (D) sodium hydroxide.
43. Which of the following is normally used for drying ammonia gas? (A) Concentrated tetraoxosulphate (VI) acid (B) Calcium oxide (C) Anhydrous calcium chloride (D) Phosphorous (V) oxide.
44. The following metals are extracted by the electrolytic method except (A) potassium (B) calcium (C) tin (D) magnesium.
45. Nuclear reactions can be used in the following except (A) gauging the thickness of objects (B) making atomic bombs (C) curing cancer (D) purifying water.
46. To what temperature must a gas be raised from 273K in order to double both its volume and pressure? (A) 298K (B) 300K (C) 546K (D) 1092K
47. Which of the following is not a direct petroleum product? (A) Ethanol (B) Petrol (C) Methane (D) Vaseline.
48. Compounds that have the same molecular formula but different structures are said to be (A) allotropic (B) isomeric (C) isotropic (D) polymeric.
49. In which of the following crystals are the particles held by Vander Waal's forces only? (A) sodium chloride (B) Ice (C) Diamond (D) Iodine.
50. Hydrogen is used for the following except (A) manufacture of ammonia (B) synthesis of hydrochloric acid (C) extinguishing fire (D) manufacture of margarine.
51. Under high pressure, real gases do not obey gas laws because their molecules.
(A) have become more energetic (B) have become less energetic (C) have become smaller in size (D) decompose into atoms.
52. Which of the following silver halides is readily soluble in ammonia hydroxide? (A) AgI (B) AgBr (C) AgCl (D) AgF.
53. Hardening of an oil may be achieved by (A) boiling the oil with water (B) boiling the oil with a catalyst (C) treating the oil with hydrogen and a catalyst (D) distilling the oil under reduced pressure.
54. Starch can be converted to ethyl alcohol by (A) distillation (B) fermentation (C) cracking (D) isomerization.
55. The oxidation state of manganese in potassium permanganate is (A) +5 (B) +3 (C) +7 (D) +2.
56. Which of the following substances are all made by the process of polymerization? (A) Nylon and soap (B) Ethanoic acid, margarine and ethanol (C) Nylon and artificial rubber (D) Nylon and margarine.
57. Zinc oxide is a (A) basic oxide (B) acidic oxide (C) amphoteric oxide (D) neutral oxide
58. When sodium chloride and metallic sodium are each dissolved in water (A) both processes are endothermic (B) both processes are exothermic (C) the dissolution of metallic sodium is exothermic (D) the dissolution of metallic sodium is endothermic.
59. The oxidation state of chlorine in potassium chlorate is (A) +1 (B) +2 (C) +3 (D) +5.
60. The nucleus of an atom contains (A) protons only (B) neutrons only (C) protons and neutrons (D) neutrons and electrons.
61. Which of the following is not present in the nucleus of a cell? (A) chromosomes (B) Nucleolus (C) Mitochondrion (D) Genes.
62. In aerobic respiration, oxidative phosphorylation takes place in the (A) cytoplasm (B) mitochondria (C) Lysosome (D) ribosomes.
63. The ability of organisms to maintain a constant internal environment is known as (A) dieresis (B) endosmosis (C) plasmolysis (D) homeostasis.
64. The bicuspid valve is located between the (A) left auricle and left ventricle (B) aorta and the left ventricle (C) superior vena cava and the inferior vena cava (D) right auricle and right ventricle.
65. The pulmonary artery carries (A) oxygenated blood from the right ventricle to the lungs (B) deoxygenated blood from the left ventricle to the right auricle (C) deoxygenated blood from the right ventricle to the lungs (D) oxygenated blood from the lungs to the right auricle.
66. Which of the following organs is responsible for controlling the body temperature regulation and water balance in mammals? (A) kidney (B) Hypothalamus (C) Parathyroid (D) Adrenal.
67. Deamination is the process during which (A) excess amino acid is broken down into ammonia and a keto-acid (B) fats and oils are converted into an emulsion for absorption (C) excess amino acid is converted into proteins (D) excess glucose is converted and stored as glycogen in the liver.
68. Which of the following is not classified as a terrestrial habitat? (A) Forest (B) Guinea savannah (C) Desert (D) Littoral Zone.
69. The relationship between a herbivore and the bacteria which live in its caecum is known as (A) parasitism (B) symbiosis (C) commensalism (D) saprophytism.
70. During blood transfusion, agglutination may occur as a result of the reaction between (A) similar antigens and antibodies (B) contrasting antigens and antibodies (C) two different antigens (D) two different antibodies.
71. The brain and the spinal cord make up the (A) automatic nervous system (B) sympathetic nervous system (C) ventral nervous system (D) parasympathetic nervous system.
72. Auxins are produced in the (A) petiole of leaves (B) apical regions of roots and shoots (C) epidermis of roots and shoots (D) parenchyma of roots and shoots.

73. The pituitary gland is called the master gland because (A) it is located in the brain (B) its secretions are more numerous than those of any other gland (C) its secretion controls other glands (D) it regulates muscular movements.
74. The pigment in the malpighian layer responsible for skin colouration is known as (A) haemoglobin (B) melanin (C) haemocyanin (D) haemoerythrin.
75. Which of the following is not an evidence of evolution? (A) Fossil (B) Anatomy (C) Embryology (D) Behaviour.
76. Which of the following is not true about gene mutation? It (A) introduced new trails into a population (B) causes changes in the DNA (C) always affects the chromosome number (D) is a source of new genes.
77. Species can be defined as the group of organisms that (A) resemble each other and live in the same habitat (B) are of a common origin and are always found together (C) resemble each other and can interbreed freely (D) resemble each other and occupy the same niche.
78. The change in colour of the chameleon serves as a means of (A) attraction to the opposite sex (B) repulsion of the enemy (C) a camouflage from a predator (D) production of a variety of skin pigments.
79. Increasing specialization of structure and function among animals of the same class or order for survival in new and different environments is called (A) adaptive radiation (B) homology (C) phylogeny (D) ontogeny.
80. The haploid number of chromosomes in man is (A) 48 (B) 46 (C) 24 (D) 23.
81. A cylinder of base radius 4cm is open at one end. If the ratio of the area of its base to that of its curved surface is 1:4, calculate the height of the cylinder. (A) 1cm (B) 2cm (C) 4cm (D) 8cm.
82. A headmaster contributes 7% of his income into a fund and his wife contributes 4% of her income, if the husband earns N5,500 per annum and the wife earns N4,000 per annum, find the sum of their annual contribution to the fund. (A) N 1.054 (B) N545 (C) N440 (D) N595.
83. A car is traveling at an average speed of 80km/hr. its speed in metres per second is (A) 13.3m/s (B) 22.2m/s (C) 133.3m/s (D) 222.2m/s.
84. Which of the following is not a measure of dispersion? (A) Mode (B) Range (C) Mean deviation (D) Standard deviation.
85. Convert 77 to a number in base two (A) 111001 (B) 1001101 (C) 100110 (D) 10101.
86. What must be added to the expression $x^2 - 18x$ to make it a perfect square? (A) 3 (B) 9 (C) 72 (D) 81.
87. The area of a parallelogram is 513cm² and the height is 19cm. Calculate the base. (A) 13.5cm (B) 25cm (C) 27cm (D) 98cm.
88. Calculate the total surface area of a cone of height 12cm and base radius 5cm. (A) $180\frac{3}{7}\text{cm}^2$ (B) $46\frac{2}{7}\text{cm}^2$ (C) $282\frac{6}{7}\text{cm}^2$ (D) $235\frac{5}{7}\text{cm}^2$
89. The area and a diagonal of a rhombus are 60cm² and 12cm, respectively. Calculate the length of the other diagonal. (A) 5cm (B) 6cm (C) 10cm (D) 12cm.
90. The angle of a sector of a circle radius 10.5cm is 120°. find the perimeter of the sector. (A) 22cm (B) 33.5cm (C) 43cm (D) 66cm.
91. The sum of the interior angles of a regular polygon is 30 right angles. How many sides has the polygon? (A) 34 sides (B) 30 sides (C) 17 sides (D) 15 sides.
92. If the exterior angles of a quadrilateral are y° , $(2y+5)^\circ$, $(y+15)^\circ$ and $(3y-10)^\circ$, find y° . (A) 61.43 (B) 60 (C) 31.43 (D) 50.
93. The sum of the angles of a regular polygon is 2520°. How many sides does the polygon have? (A) 20 (B) 18 (C) 15 (D) 16.
94. Which of the following is not a property of all parallelograms? (A) The diagonals bisect each other (B) the diagonals cut at right angles (C) The opposite sides are equal (D) The opposite angles are equal.
95. The length of an exercise book is given as 20cm correct to the nearest centimeter. In which of the following ranges of possible measurements does the actual length lie? (A) 19.1 - 20.9cm (B) 19 - 20.5cm (C) 19.6 - 20.5cm (D) 19.9 - 20.1cm.
96. If 3 children share N 10.50 among themselves in the ratio 6:7:8, how much is the largest share? (A) N3.00 (B) N2.00 (C) N3.50 (D) N5.00.
97. By selling some crates of soft drinks for N600.00, a dealer makes a profit of 50%. How much did the dealer pay for the drinks? (A) N400.00 (B) N450.00 (C) N200.00 (D) N900.00.
98. A hollow sphere has a volume of $k\text{cm}^3$ and a surface area of $k\text{cm}^2$. Calculate the diameter of the sphere (A) 3cm (B) 6cm (C) 9cm (D) More information is needed.
99. Find the median of the following numbers: 2.64, 2.50, 2.72, 2.91 and 2.35 (A) 2.92 (B) 2.72 (C) 2.64 (D) 2.50.
100. A baker used 40% of a 50kg bag of flour. If $\frac{1}{8}$ of the amount used was for cake, how many kilogrammes of flour was used for cake? (A) $21\frac{1}{2}$ (B) $61\frac{1}{4}$ (C) $15\frac{1}{8}$ (D) $17\frac{1}{2}$

ANSWERS TO NAU POST-LIME SCREENING EXERCISE (2015 / 2016)
SESSION FOR MEDICINE, NATURAL SCIENCES, HEALTH SCIENCES AND
TECHNOLOGY

1. D	2. B	3. B	4. D	5. C	6. A	7.D	8.B	9.C	10.B	11.D	12.A
13.A	14.B	15.D	16.B	17.B	18.B	19.D	20.C	21.C	22.A.	23.D	24.D
25.B	26.B	27.	28.B	29.B	30.D	31.C	32.B	33.A	34.D	35.C	36.
37.A	38.C	39.C	40.B	41.	42.C	43.	44.B	45.C	46.D	47.D	48. B
49. B	50. C	51. A	52. B	53. D	54. B	55. B	56. C	57. A	58. C	59. B	60. C
61.	62. D	63.	64.	65.	66.	67.	68. D	69. A	70.	71. C	72.
73. A	74. B	75. A	76. B	77. A	78. C	79.	80.				



**POST-UTME SCREENING EXERCISE, 2016/2017 FOR
MEDICINE, BASIC MEDICAL SCIENCES, HEALTH
SCIENCES AND TECHNOLOGY, PHARMACEUTICAL
SCIENCES, ENGINEERING,
BIOSCIENCES, PHYSICAL SCIENCES,
ENVIRONMENTAL SCIENCES, AGRICULTURE.**

INSTRUCTION

1. Shade FIRMLY only one answer using HB pencil only
2. You are allowed to make a second choice of course into the following: Faculty of Agriculture, Geophysics, Anatomy, Science Education, Federal College of Education (Technical) Umunze, Enugu State College of Education (Technical). (See the back of the OMR sheet for details)
3. Shade your second choice of course in the column provided (Column 8) Time allowed: 1 hour:

Date: 6th August, 2011

Fill in blank in the following sentences making use of the best of the options.

1. The project which seemed very near realization Because of lack of funds. A. fell over B. fell through C. fell off D. fell in E. fell down
2. The members of the panel were A. discussing about it B. discussing on it C. discussing upon it D. discussing it E. discussing around it
3. It is time we A. should be going B. ought to go C. are going D. went E. go
4. The way to stop some frivolous publication is to The press A. gag B. shackle C. fetter D. handcuff E. blind
5. Without Words he accused him directly of treachery. A. amending B. modifying C. mixing D. mincing E. meaning

Choose the word or phrase closest in meaning to the underlined world.

6. Most times my classmate tells cock and bull stories. A. costly B. unreasonable C. clever D. untrue E. funny
7. Never in the history of human conflict has so much been owed by so many to so few. A. many people owed much money at the end of the war. B. A handful of people saved the lives of a nation C. A few people did a lot of things gratis. D. This conflict caused the largest ransom ever demanded E. Very little was owed by anyone to anybody
8. The ingenious scheme of the robber broke down because of the efficiency of the police officer Ingenious here means. A. suspicious B. ambitious C. deceitful D. clever E. stupid
9. Be careful we are approaching the crossroads. A. look up B. watch out C. look out D. hold up E. look
10. He took exception to Njoku's remark A. was delighted B. was excited by C. got demoralized by D. objected to E. was perplexed by

CHOOSE THE OP7TON OPPOSITE IN MEANING TO THE WORD(S) ORPHRASE UNDERLINED

11. For millions of years, the world's resources have remained boundless. A. unlimited B. scarce C. indomitable D. limited E. available
12. The president took exception to the ignoble role the young man played in the matter. A. honourable B. embarrassing C. dishonourable D. extraordinary E. dishonest
13. He accepted a mundane task without hesitation. A. great B. lowly C. menial D. moderate E. ordinary
14. In a sentence, there must always be concord between a subject and it verb. A. breath B. departure C. disagreement D. dispute E. alliance
15. The lecture was merely obfuscating the issue with his endless examples. A. complicating B. clarifying C. confusing D. summarizing E. boring

FILL EACH GAP WITH THE MOST APPROPRIATE OPTION FROM THE LISTFOLLOWING THE GAP.

16. My shoes are worn out; I must see my A. grocer B. tanner- C. cobber D. hosier E. shoe-maker
17. A. A 90-years B. A 90-year old C. a 90 year-old D. 90-year old E. 90-years old) woman was attacked by a group of bandits.
18. She has been here (A. since B. for C. from D. n E. during) the past three weeks.
19. Before mechanization, workers (A. wring B. wrung C. wrang D. wringed E. wrong) water out of fabrics with their hands.
20. Three quarters of the physics class A. improve B. improves C. are improving D. is improving E. is improved) dramatically.
21. Eze is contemptuous (A. to B. at C. for D. of E. from) dishonest people.
22. The plaintiff (A. asked B. begged C. demanded of D. prayed E. forced) the court to restrain the defendants from further action.
23. Ngozi is prepared for a . (A: show-off B. show-down C. show-out D. show-up E. show-up) with his opponent following his defeat last season.
24. The President rejected the bill and withheld his (A. accent B. assent C. access D. ascent E. assess)
25. The dark continent was an (A. abuse B. enigma C. eclectic D. epithet E. impression) for Africa.

CHOOSE THE BEST OF THE OPTIONS FROM THE QUESTIONS BELOW

26. Which of the following metals cannot replace hydrogen from water or steam?
A. Sodium B. Magnesium C. Iron D. Calcium E. Copper
27. Which one of the following change is physical? A. adding iron filings to aerated water B. adding sodium metal to water C. cooling a solution of iron(II) sulphate to obtain the hydrate salt. D. cooling water to obtain ice E. Buring the domestic gas for cooking
28. 0.1 Faraday of electricity was passed through a solution of copper (II) sulphate. The maximum weight of copper deposited on the cathode would be (Cu=64) A. 64.0g B. 32.0g C. 16.0g D. 6.4g E. 3.2g
29. Bronze is an alloy of A. copper, zinc and nickel B. aluminum and copper C. copper and zinc D. tin and zinc E. tin and copper
30. Starch can be converted to ethyl alcohol by A. distillation B. fermentation C. isomerization D. cracking E. cooking

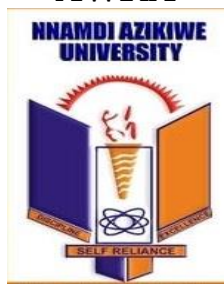
31. Pick out the correct statement A. zinc heated to redness reacts with steam to give oxygen and zinc oxide B. zinc heated to redness reacts with steam to give oxygen and hydrogen. C. zinc does not react with hot or cold water D. zinc reacts with hot water to form zinc oxide and hydrogen E. zinc reacts very easily with cold water to give zinc oxide and hydrogen.
32. Chlorine, bromine and iodine resemble each other in that all A. dissolve in alkalis B. react violently with hydrogen without heating C. displace each other from solutions of their salts D. are gases E. are liquids
33. When air is compressed A. its density decreases B. it solidifies c. its temperature decreases D. its temperature increases E. its temperature unchanged.
34. Which of the following is formed when maltose acts with concentrated tetraoxosulphate (VI) acid? A. Carbon (IV) oxide B. Coal tar C. Charcoal D. Toxic fumes E. None of the above.
35. A property of a colloidal dispersion which a solution does not have is A. the Tyndall effect B. homogeneity C. osmotic pressure D. surface polarity E. all of the above.
36. Dr Nnamdi Azikiwe was born in A. 1909 B. 1900 C. 1904 D. 1901 E. 1902
37. One of the following was a female Governor in Nigeria. A. Prof. Dora Akunyili B. Dame Patience Goodluck C. Hon. Uche Ekwunife D. Dame Vergy Etiaba E. Chief (Mrs.) Serah Jubril
38. The 2010 soccer world-cup competition took place in A. Germany B. Brazil C. Colombia D. USA E. South Africa

ANSWERS

- | | | | | | |
|------|------|------|------|------|------|
| 1.D | 2.B | 3.D | 4.A | 5.D | 6.D |
| 13.A | 14.C | 15.B | 16.C | 17.A | 18.B |
| 25.D | 26. | 27.D | 28.A | 29.E | 30.B |
| 37.D | 38.E | 39.D | 40.C | 41.A | 42.A |
| 49.C | 50.C | | | | |

39. The first Nigerian Vice-Chancellor was? A. Prof Eni Njoku B. Prof Takenna Tamuno C. Prof Ade Ajayi D. Prof. Kenneth Dike E. Prof. Adeoye Lambo
40. The capital of Lagos State is A. Surulere B. Victoria Island C. Ikeja D. Oshodi E. Ikorodu
41. If x is positive for what range of x is $4 + 3x < 10$? A. $0 < x < 2$ B. $x < x$ C. $1 < x < 2$ D. $0 < x < 4\frac{3}{4}$ E. $0 > x > 2$
42. A house bought for #109,000 was later auctioned for #80,000. Find the loss percent. A. 20% B. 30% C. 40% D. 50% E. 60%
43. If $2x + y = 7$ and $3x - 2y = 3$, by how much is 7x greater than 10? A. 1 B. 3 C. 7 D. 10 E. 17
44. Find the number whose logarithm to base 10 is 2.6025. A. 400.4 B. 0.4004 C. 0.04004 D. 0.004004 E. 0.0004004
45. Simplify: $\frac{3}{4} + 1\frac{1}{4} \times (1\frac{1}{2} - \frac{2}{3})$. A. $\frac{7}{30}$ B. $\frac{7}{24}$ C. $\frac{9}{25}$ D. $\frac{1}{2}$ E. $\frac{18}{25}$
46. A stick of length 1.7M was measured by a boy as 1.80M. find the percentage error in his measurement. A. 27%/9 B. 26%/7 C. 5% D. 27 7%/9 E. 28 4%/7
47. Simplify $\log_3 9 + \log_3 15 - \log_3 5$. A. $\log_3 19$ B. $\log_3 3$ C. 3 D. 1 E. 0
48. A trader makes a loss of 15% when selling an article. Find the ratio, selling price: cost price. A. 3:20 B. 3: 17 C. 17:20 D. 20: 23 E. 3:23
49. If N varies directly as M and N = 8 when M = 20. Find M when N = 7. A. 13 B. 15 C. $17\frac{1}{2}$ D. $18\frac{1}{2}$ E. $19\frac{1}{2}$
50. Solve for T in the equation $\frac{3}{4}t + \frac{1}{3}(12 - t) = 11$. A. $\frac{9}{13}$ B. $3\frac{9}{13}$ C. 5 D. $9\frac{3}{5}$ E. $5\frac{9}{13}$

- | | | | | | |
|------|------|------|------|------|------|
| 7.B | 8.D | 9.C | 10.D | 11.D | 12.A |
| 19.A | 20.B | 21.A | 22.C | 23.B | 24.B |
| 31.A | 32.C | 33.C | 34.A | 35.A | 36.C |
| 43.C | 44.C | 45.B | 46.C | 47.C | 48.C |



POSTOUME SCREENING EXERCISE, 2017/2018

FOR

MEDICINE, BASIC MEDICAL SCIENCES, HEALTH SCIENCES AND TECHNOLOGY, PHARMACEUTICAL SCIENCE, BIOSCIENCES AND AGRICULTURE.

INSTRUCTION:

1. Shade FIRMLY only one answer using HB pencil only
2. You can allowed to make a second choice of course into the following:
faculty of Agriculture, Geophysics, Anatomy, science Education, Federal College of Education (Technical) Umunze, Enugu State college of Education (Technical). (see the back of the DMR sheet for details)
3. Shade your second choice of course in the column provided (Column 8)

Time allowed: 1 hour

Date: 30th June, 2012

Fill in the following sentences making use of the best of the options.

1. Njoku.....as a mechanic when he was young, but now he is a driver. A. Had been working B. used to work C. Would work D. would have worked E. had worked.
2. If Ikenna performs poorly in her forthcoming examination, then either her mother or I to blame. A Must B. has C. will be D. are E. am
3. No sooner had the examination ended the students started vacating the halls. A. When B. that C. than D. before E. and
4. f are day we would talk about it for weeks. A. a lion would be coming into this room B. a lion was coming into this room C. a lion came into this room D. a lion is coming into this room E. a lion is going to come into this room
5. When we are coming back from the picnic last Friday there on Niger Bridge. A. Were many traffics B. was much traffic C. Were plenty of traffic D. were lots of traffic

Choose the word or phrase closest in meaning to the underlined word or phrase.

6. My friend will hate his uncle for ever because he left him in the lurch in his hour of need. A. abandon him B. punished him C. disrespected him D. disinherited him E. despised him.
7. For all he cared, the game was as good as lost. A. He did not care if the game was lost B. He could not care less if the game was lost C. He was almost certain that the game would be lost D. He was afraid the game would be lost. E. The game was very good and he cared very much about losing it.
8. Some people are naturally belligerent. A. Naughty B. benevolent C. familiar D. clear E. crazy

9. This is a mundane thing. A. natural B. secular C. familiar D. clear E. crazy
10. The word caligynophobia means. A. fear of smile B. fear of nudity C. fear of pregnancy D. fear of beautiful women E. fear of being touched

Choose the option opposite in meaning to the word (s) or phrase underlined.

11. The man who had been seriously ill was convalescing at a sea-side resort.
A. Regaining health B. deteriorating in health C. recuperating D. relaxing E. hallucinating
12. Ben will soon get to the zenith of his career. A. Peak B. low C. Nadine D. bottom E. foot.
13. The antonym of the word verbose is? A. Lose B. concise C. intractable D. overdue E. heterodox

Choose the word/expression which best completes each sentence

14. You.....To know better than an illiterate farmer. A. Are suppose B. are supposed C. supposed D. may suppose E. will suppose
15. The passenger waited patiently whilst they transferred his.....From the' taxi to the bus. A. properties B. baggages C. baggage D. property E bagagges.

Choose the best of the options from the Questions below

16. If hydrogen sulphide gas is passed into a solution of a pure iron chloride, a yellow deposit appears. If left behind. The pale green solution is A. Dilute sulphuric acid B. dilute hydrochloric acid C. unreacted hydrogen sulphide in water D. iron (II) chloride E. iron (II) chloride
17. A solid X when heated gives off a brown gas. If X is soluble in excess sodium hydroxide solution but insoluble in excess ammonium hydroxide solution, then X isA. Basic lead carbonate B. lead (II) trioxonitrate C. Sodium carbonate D. zinc nitrate E. sodium nitrate
18. When excess ethanol is heated to 145°C in the presence of concentrated sulphuric acid, the product is. A. Diethyl ether B. ethyne C. diethyl sulphate D. acetone E. ethanoic acid
19. Given the mean atomic mass of chlorine prepared in the laboratory to be 35.5 and assuring that chlorine contains two isotopes of mass number 35 and 37, what is the percentage composition of the isotope of mass number 35? A. 20 B.25 C. 50. D. 75 E.90
20. The oxidation state of managense in KMnO_4 is. A. +1 B.+3 C.+5 D+7 E+2
21. 28.8cm^3 of nitrogen at 15°C is cooled to 0° at constant pressure, the new volume of nitrogen is. A. 17.4cm^5 B. 14.7cm^5 C. 27.3cm^3 D 31.7cm^3 E. 34.7cm^3
22. The electronics configurations of elements X and Y are X = 2,8,6 ; Y 2,8,7. The bond in the compound formed by X and Y is expected to be A. ionic B. covalent C. dartive covalent D. metallic E. none of the above
23. If 1 mole of aluminum contains 6×10^{23} atoms of aluminum, how many atoms are contained in 0.9g of aluminum? A. 1.0×10^{23} B. 6.6×10^{23} C. 2.0×10^{23} D. 6.0×10^{23} E. 5.4×10^{23}
24. 1.Og of the metal when reduced with hydrogen. The formula of the oxide is. A. X_2O B. XO C. XO_2 DX $_2\text{O}_3$ EX $_3\text{O}_4$
25. Cotton is almost pure. A. Starch B. protein C. animal fibre D.cellulose E. mineral fibre
26. A spirogyra cell was placed in solution X. After 1 minute the cell had been plasmolysed, solution X was. A. Tap water B. a very strong sugar solution C. A weak sugar solution D. A very weak sugar solution E. distilled water

27. Which of these product obtained from analyzing sweat from the skin is not an excretory product? A. Water B. salts C. dust D. urea E. urates
28. An organism which lives on the remains of the dead plant is. A. An endoparasite B. A saprophyte C. A commensal D. A symbiont An ectoparasite
29. Which of the substances listed below is a trace element for plants? A. Potassium B. sodium C. copper D. phosphate E. Nitrate
30. Which of these is not a type of soil? A. Sand B. Granite C. Loam D. Clay E. Sandy loan
31. In which of the following tissues do the cells have the ability to divide?
A. Meristem B. Sclerenchyma C. Philoem D. Collenchyma E. pith
32. Root hairs have their origin form. A. Endodermis B. pericycle C. Cortex D. epidermis E. Xylem
33. Magnesium is utilized in the formation of. A. ATP B. glucose C. amino acids D. Chlorophyll E. fats
34. Which of the following organs produces bile? A. Gall bladder B. Pancreas C. spleen D. Liver E. a Stomach
35. The following are connected with the movement of a reflex action.(1) Central nervous system (2) muscle (3) skin (4) sensory nerve (5) Motor nerve
Which of the following sequences indicates a correct path?
A. 1-2-3-4-5 B. 2-1-4-5-3 C. 3-4-1-2-5 D. 4-1-5-2-3 E. 3-4-1-2-5
36. During thunderstorm the energy of lightning discharge causes. A. Oxygen and nitrogen to combine B. More carbon dioxide to be formed C. Nitrates to be converted to nitrogen D. death
37. The amount of water loss from a leaf can be detected using. A. lime-water B. red litmus paper C. blue litmus paper D. blue cobalt chloride paper E. pink cobalt chloride paper.
38. The mushroom is an organism which feeds. A. parasitically B. saprophytically C. holozonically D. autotrophically

(ANSWERS)

- (1)B (2)D (3)A (4)A (5)B (6)A (7)A (8)B (9)A (10)D (11)B (12)C
(13)B (14). (15)B (16) E (17) B (18) A (19) D (20)D (21) C (22)B (23) C (24) A
(25) D (26) B (27) C (28) B (29) E (30) B (31) A (32) E (33) A (34) A (35) D
(36) D (37) D (38) E (39) D (40) (41) E (42) C (43) A (44) B (45) D (46) A
(47) C (48) D (49) B (50) B.

39. The tissue which connects a muscle to a bone is a. A. Cartilage B. muscle fibre C. tendon D. ligament E. connecting fibre
40. In which of the following plants is swollen shoot disease common? A. Groundnut B. Cocoa C. corn D. cotton E. rubber
41. A pyramid is constructed on a cuboid. The figure has. A. Twelve faces B. thirteen vertices C. Fourteen edges D. Fifteen edges E. sixteen edges
42. If $X^2 + 4 =$, then X. A. 4 B. -4 C.2 D.-2 E. none of these
43. $(3.2)^2 - (1.8)^2$ equals. A. 7.0 B. 2.56 C.13.48 D. 2.0 E. 0.07
44. The annual profits of a transport business were divided between the two partners A and B in the ratio 3:5. If B received \$3000 more than A, the total profit was. A.\$5000 B. \$1300 C. \$12000 D.\$24000 E.\$8000
45. Which of the following is not a factor of $12^4 - 5^4$? A.7 B.13 C.17 D.49 E.169
46. A temperature of 20°C is the same as. A. 36°F B. 68°F C. 11.1°F D. 48.1°F E. 25°F
47. A guiter string is 75cm long. The wavelength of its fundamental note is A. 75cm B. 150cm C.37.5cm D.112.5cm E. 50cm
48. A man standing between two parallel mirrors in a barber's shop will see the following number of his own image. A. Eight B. Two C. Four D. Five E. infinite
49. In an electrolysis experiment, a cathode of mass 5g is found to weigh 5.01g after a current of 5A flows for 50 seconds. What is the electrochemical equivalent of the deposited substance? A. 0.000042/C B. 0.00002g/C C. 0.02500g/C D.0.05000g/C E. 0.00001g/C
50. Which of the following electromagnetic waves has the shortest wavelength?A. Radio B. X-rays C. infra-red D. Blue light E. Ultraviolet